

Sanjeev Gautam

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PROFESSIONAL SUMMARY

Computational scientist with a Ph.D. in Computational Chemistry and expertise in molecular simulations, statistical mechanics, scientific computing, and high-performance computing, with experience applying machine learning to scientific problems. Develops predictive models and automated workflows to understand how molecular interactions and process-relevant conditions influence membrane vesicles, transport, phase behavior, and material stability. Interested in applying physics-based modeling and AI/ML to biomanufacturing, formulation development, and therapeutic delivery.

TECHNICAL SKILLS

- **Modeling & Simulation:** Statistical mechanics, thermodynamic modeling, coarse-grained and atomistic molecular dynamics, model validation, parameter-space exploration
- **Membranes & Soft Matter:** Lipid vesicles, membrane mechanics, osmotic stress, molecular transport, phase separation, interfacial behavior, self-organization
- **Programming & Data Analysis:** Python (NumPy, SciPy, pandas), C, Bash; trajectory analysis, statistical inference, workflow automation, data processing, and visualization
- **Machine Learning:** scikit-learn, XGBoost; feature engineering, classification, clustering, predictive modeling, and model interpretation
- **High-Performance Computing:** Linux/Unix, Slurm, HTCondor, MPI; distributed simulations, high-throughput studies, and reproducible workflows
- **Software:** LAMMPS, GROMACS, NAMD, OpenMM, VMD
- **Computational Chemistry:** Density Functional Theory (DFT), electronic structure calculations, intermolecular interactions, and spectroscopy-guided interpretation

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher

Oct 2025 – Present

Kansas State University, Manhattan, KS

- Develop predictive computational models combining solution thermodynamics, membrane mechanics, and molecular simulations to identify conditions influencing vesicle stability, loading efficiency, and formulation performance.
- Evaluate how process and formulation variables affect vesicle behavior during loading, buffer exchange, storage, and therapeutic delivery applications.
- Model phase separation, wetting, and interfacial behavior in multicomponent soft materials to determine how composition and molecular interactions control morphology and stability.
- Develop automated Python and HPC workflows for simulation setup, parameter exploration, trajectory processing, quantitative analysis, and data visualization.
- Apply statistical and machine-learning methods to classify material behavior, identify key performance drivers, and map structure–property relationships.
- Translate computational results into technical reports, presentations, and visual materials for interdisciplinary research and industry-oriented audiences.

Doctoral Researcher

Aug 2017 – Apr 2025

University of Pittsburgh, Pittsburgh, PA

- Developed coarse-grained molecular dynamics models to determine how molecular interactions, crowding, and spatial organization regulate selective transport through complex biomolecular environments.
- Quantified molecular diffusion, transport pathways, spatial distributions, and interaction patterns across multiple biological conditions.
- Designed and executed large-scale MPI-enabled simulation studies across diverse molecular compositions and interaction conditions on multi-node HPC systems.
- Created Python tools for trajectory processing, transport measurements, interaction mapping, statistical characterization, and visualization of large simulation datasets.
- Evaluated competing molecular mechanisms by comparing simulations with theoretical expectations and published experimental observations.
- Collaborated across computational chemistry, biophysics, and molecular simulation and communicated findings through peer-reviewed publications and scientific presentations.

Graduate Teaching Assistant

Aug 2017 – Apr 2020

University of Pittsburgh, Pittsburgh, PA

- Taught chemistry laboratory courses and mentored students in experimental methods, quantitative analysis, thermodynamics, computational chemistry, and scientific problem-solving.

Master's Researcher and Summer Research Intern

May 2014 – Jun 2016

NISER; Indian Institute of Science; IGCAR, India

- Applied Density Functional Theory and spectroscopic analysis to investigate intermolecular interactions, molecular conformations, and structure–property relationships relevant to molecular design and materials characterization.
- Compared computational predictions with condensed- and gas-phase spectroscopic measurements to interpret molecular structure and connect predicted properties with experimental observations.

SELECTED PROJECT

- **EGFR Inhibitor ML Pipeline** |  (github.com/skg43/EGFR_RF_Modeling) – Built an end-to-end Python/RDKit pipeline for molecular descriptor and fingerprint generation, chemical similarity analysis, Random Forest and XGBoost model development, and virtual screening of EGFR inhibitors.

EDUCATION

University of Pittsburgh

Aug 2017 – Apr 2025

Ph.D., Computational Chemistry

NISER, India

Aug 2011 – Jun 2016

Integrated M.Sc., Physical Chemistry

PUBLICATIONS

- R. G. Pereira, B. Mukherjee, **S. Gautam**, M. D'Agnese, S. Biswas, R. Meeker, and B. Chakrabarti, *Osmotically Induced Shape Changes in Membrane Vesicles*, bioRxiv (2026). [doi:10.64898/2026.04.03.716363](https://doi.org/10.64898/2026.04.03.716363).
- **S. Gautam**, R. Laghaei, A. Eskandari Nasrabad, and R. D. Coalson, *Computational Studies of Cargo Transport through the Nuclear Pore Complex*, bioRxiv (2026). [doi:10.64898/2026.02.23.707554](https://doi.org/10.64898/2026.02.23.707554).
- V. Mundlapati, **S. Gautam**, D. Sahoo, A. Ghosh, H. S. Biswal, *Thioamide, a Hydrogen Bond Acceptor in Proteins and Nucleic Acids*, J. Phys. Chem. Lett. **8**, 4573–4579 (2017).